

9.4 Refer to Exercise 9.3. For each of the following pairs of contrasts, determine whether they are orthogonal.

- Question 1 and question 2
- Question 1 and question 3
- Question 1 and question 4
- Question 2 and question 3
- Question 2 and question 4
- Question 3 and question 4
- Are the four contrasts mutually orthogonal?

9.5 Refer to Example 8.4. The researchers were interested in determining whether the mean oxygen content was lower for samples taken near the mouth of the Mississippi than for samples taken further away from the mouth. Write a contrast to answer each of the following questions and test whether the contrast is different from zero using $\alpha = .05$. Clearly summarize your results.

- Is the mean oxygen content at 20 KM different than the average of the mean oxygen content at 1 KM, 5 KM, and 10 KM?
- Is the mean oxygen content at 10 KM different than the average of the mean oxygen content at 1 KM and 5 KM?
- Is the mean oxygen content at 5 KM different than the average of the mean oxygen content at 1 KM?
- Are the three contrasts defined in (a), (b), and (c) mutually orthogonal?
- Do the three contrast sums of squares total to SS_{Tr} ?

9.3 Which Error Rate Is Controlled?

individual comparisons

experimentwise error rate

An experimenter wishes to compare t population (treatment) means using m contrasts. Each of the m contrasts can be tested using the F test we introduced in the previous section. Suppose each of the contrasts is tested with the same value of α , which we will denote as α_I , called the **individual comparisons** Type I error rate. Thus, we have an α_I chance of making a Type I error on each of the m tests. We need to also consider the probability of falsely rejecting at least one of the m null hypotheses, called the **experimentwise** Type I error rate and denoted by α_E . The value of α_E takes into account that we are conducting m tests, each having an α_I chance of making a Type I error. Now, if MS_{Error} has an infinite number of degrees of freedom (so the tests are independent), then when all m null hypotheses are true, the probability of falsely rejecting at least one of the m null hypotheses can be shown to be $\alpha_E = 1 - (1 - \alpha_I)^m$. Table 9.4 contains values

TABLE 9.4

A comparison of the experimentwise error rate α_E for m independent contrasts among $t(t > m)$ sample means

m , Number of Contrasts	α_I Probability of a Type I Error on an Individual Test		
	.10	.05	.01
1	.100	.050	.010
2	.190	.097	.020
3	.271	.143	.030
4	.344	.185	.039
5	.410	.226	.049
⋮	⋮	⋮	⋮
10	.651	.401	.096

of α_E for various values of m and α_I . We can observe from Table 9.4 that as the number of tests m increases for a given value of α_I , the probability of falsely rejecting H_0 on at least one of the m tests α_E becomes quite large. For example, if an experimenter wanted to compare $t = 20$ population means by using $m = 10$ orthogonal contrasts, the probability of falsely rejecting H_0 on at least one of the t tests could be as high as .401 when each individual test was performed with $\alpha_I = .05$.

In any practical problem, the degrees of freedom for MS_{Error} will not be infinite and hence the tests will not be independent. Thus, the relationship between α_E and α_I is not generally as described in Table 9.4. It is difficult to obtain an expression equivalent to $\alpha_E = 1 - (1 - \alpha_I)^m$ for comparisons made with tests that are not independent. However, it can be shown that for most of the types of comparisons we will be making among the population means, the following upper bound exists for the experimentwise error rate:

$$\alpha_E \leq 1 - (1 - \alpha_I)^m$$

Thus, we know the largest possible value for α_E when we set the value of α_I for each of the individual tests. Suppose, for example, that we wish the experimentwise error rate for $m = 8$ contrasts among $t = 20$ population means to be at most .05. What value of α_I must we use on the m tests to achieve an overall error rate of $\alpha_E = .05$? We can use the previous upper bound to determine that if we select

$$\alpha_I = 1 - (1 - \alpha_E)^{1/m} = 1 - (1 - .05)^{1/8} = .0064$$

then we will have $\alpha_E \leq .05$. The only problem is that this procedure may be very conservative with respect to the experimentwise error rate, and hence an inflated probability of Type II error may result.

Bonferroni inequality

We will now consider a method that will work for any set of m tests and is much easier to apply in obtaining an upper bound on α_E . The results of Table 9.4 are disturbing when we are conducting a number of tests. The chance of making at least one Type I error may be considerably larger than the selected individual error rates. This could lead us to question significant results when they appear in our analysis of experimental results. The problem can be alleviated somewhat by *controlling the experimentwise error rate* α_E rather than the *individual error rate* α_I . We need to select a value of α_I that will provide us with an acceptable value for α_E . The **Bonferroni inequality** provides us with a method for selecting α_I so that α_E is bounded below a specified value. This inequality states that the overall Type I error rate α_E is less than or equal to the sum of the individual error rates for the m tests. Thus, when each of the m tests have the same individual error rates α_I , the Bonferroni inequality yields

$$\alpha_E \leq m\alpha_I$$

If we wanted to guarantee that the chance of a Type I error was at most α , we could select

$$\alpha_I = \frac{\alpha}{m}$$

for each of the m tests. Then,

$$\alpha_E \leq m\alpha_I = m \left(\frac{\alpha}{m} \right) = \alpha$$

The experimentwise error rate is thus less than or equal to our specified value. Just as we mentioned earlier, this procedure may be very conservative with respect to the experimentwise error rate, and hence an inflated probability of Type II error may result.

EXAMPLE 9.5

Refer to Example 9.4, where we constructed $m = 4$ contrasts (comparisons) among the $t = 5$ treatment means. If we wanted to control the experimentwise error rate at a level of $\alpha_E = .05$, then we would take

$$\alpha_i = \frac{0.5}{4} = .0125$$

The critical value for the F tests would then be $F_{.0125, 1, 25} = 7.24$ as opposed to $F_{.05, 1, 25} = 4.24$ if we ignore the fact that we are conducting multiple tests on the treatment means. We would then reject H_0 if $SSC_i/MS_{\text{Error}} \geq 7.24$. From Example 9.4, the four F ratios were

$$F_1 = 13.71 \quad F_2 = 8.48 \quad F_3 = 0.24 \quad F_4 = 1.42$$

Using the Bonferroni procedure, we would declare contrast l_1 and l_2 significantly different from 0 because their F ratios are greater than 7.24. Using the Bonferroni test procedure, we are assured that the chance of making at least one Type I error during the four tests is at most .05. Using $\alpha = .05$ for each of the four procedures would not have allowed us to assess the exact probability of making a Type I error among the four comparisons. However, this value would have been considerably larger than .05, possibly as large as .20.

The Bonferroni procedure gives us a method for evaluating a small number of contrasts that were selected prior to observing the data while preserving a selected experimentwise Type I error rate. In many experiments, the researcher will want to compare all pairs of treatments or compare all treatments to a control. In these situations, there are many methods for testing these types of contrasts among the treatment means. A major difference among these multiple-comparison procedures is the type of error rate that each procedure controls. We will discuss several of these procedures in the next sections.

9.4 Fisher's Least Significant Difference

Recall that we are interested in determining which population means differ after we have rejected the hypothesis of equality of t population means in an analysis of variance. R. A. Fisher (1949) developed a procedure for making pairwise comparisons among a set of t population means. The procedure is called Fisher's least significant difference (LSD).

The α -level of Fisher's LSD is valid for a given comparison only if the LSD is used for independent (orthogonal) comparisons or for preplanned comparisons. However, since many people find Fisher's LSD easy to compute and hence use it for making all possible pairwise comparisons (particularly those that look "interesting" following the completion of the experiment), researchers recommend applying Fisher's LSD only after the F test for treatments has been shown to be significant. This revised approach is sometimes referred to as **Fisher's protected**

Fisher's protected LSD

LSD. Simulation studies [Cramer and Swanson (1973)] suggest that the error rate for the protected LSD is controlled on an experimentwise basis at a level approximately equal to the α -level for the F test.

We will illustrate Fisher's protected procedure, but will continue to call it Fisher's LSD. This procedure is summarized here.

Fisher's Least Significant Difference Procedure

1. Perform an analysis of variance to test $H_0: \mu_1 = \mu_2 = \cdots = \mu_t$ against the alternative hypothesis that at least one of the means differs from the rest.
2. If there is insufficient evidence to reject H_0 using $F = \text{MSB}/\text{MSW}$, proceed no further.
3. If H_0 is rejected, define the **least significant difference (LSD)** to be the observed difference between two sample means necessary to declare the corresponding population means different.
4. For a specified value of α , the least significant difference for comparing μ_i to μ_j is

$$\text{LSD}_{ij} = t_{\alpha/2} \sqrt{s_W^2 \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

where n_i and n_j are the respective sample sizes from population i and j and t is the critical t value (Table 2 of the Appendix) for $\alpha = \alpha/2$ and df denoting the degrees of freedom for s_W^2 . Note that for $n_1 = n_2 = \cdots = n_t = n$,

$$\text{LSD}_{ij} \equiv \text{LSD} = t_{\alpha/2} \sqrt{\frac{2s_W^2}{n}}$$

5. Then compare all pairs of sample means. If $|\bar{y}_i - \bar{y}_j| \geq \text{LSD}_{ij}$, declare the corresponding population means μ_i and μ_j different.
6. For each pairwise comparison of population means, the probability of a Type I error is fixed at a specified value of α .

Note: The LSD procedure is analogous to a two-sample t -test for any two population means μ_i and μ_j , except that we use s_W^2 , the pooled estimator of the population variance σ^2 from all t samples, rather than the pooled sample variance from samples i and j . Also, the degrees of freedom for the t value is $\text{df}_W = n_T - t$ from the analysis of variance, rather than $n_i + n_j - 2$.

EXAMPLE 9.6

Refer to Example 9.3, where we had five different weed agents and $n = 6$ plots of land assigned to each of the agents. The analysis of variance is given in Table 9.5.

TABLE 9.5
AOV table for the
data of Example 9.3

Source	df	SS	MS	F	p -value
Treatment	4	.3648	.0912	5.96	.0016
Error	25	.3825	.0153		
Totals	29	.7472			

steps for LSD procedure

Solution We can solve this problem by following the five steps listed for the LSD procedure.

Step 1. We use the AOV table in Table 9.5. The F test of $H_0: \mu_1 = \mu_2 = \dots = \mu_5$ is based on

$$F = \frac{MSB}{MSW} = \frac{MS_{\text{Trit}}}{MS_{\text{Error}}} = 5.96$$

For $\alpha = .05$ with $df_1 = 4$ and $df_2 = 25$, we reject H_0 if F exceeds 2.76 (see Table 8 in the Appendix).

Steps 2, 3. Since 5.96 is greater than 2.76, we reject H_0 and conclude that at least one of the population means differs from the rest ($p = .0016$).

Step 4. The least significant difference for comparing two means based on samples of size 6 is then

$$LSD = t_{\alpha/2} \sqrt{\frac{2 MS_{\text{Error}}}{6}} = 2.060 \sqrt{\frac{2(.0153)}{6}} = .1471$$

Note that the appropriate t value (2.060) was obtained from Table 2 with $a = \alpha/2 = .025$ and $df = 25$.

Step 5. When we have equal sample sizes, it is convenient to use the following procedures rather than make all pairwise comparisons among the sample means, because the same LSD is to be used for all comparisons.

a. We rank the sample means from lowest to highest.

Agent	1	2	3	4	5
$\bar{y}_{i.}$	1.175	1.293	1.328	1.415	1.500

b. We compute the sample difference

$$\bar{y}_{\text{largest}} - \bar{y}_{\text{smallest}}$$

If this difference is greater than the LSD, we declare the corresponding population means significantly different from each other. Next we compute the sample difference

$$\bar{y}_{\text{2nd largest}} - \bar{y}_{\text{smallest}}$$

and compare the result to the LSD. We continue to make comparisons with $\bar{y}_{\text{smallest}}$:

$$\bar{y}_{\text{3rd largest}} - \bar{y}_{\text{smallest}}$$

and so on, until we find either that all sample differences involving $\bar{y}_{\text{smallest}}$ exceed the LSD (and hence the corresponding population means are different) or that a sample difference involving $\bar{y}_{\text{smallest}}$ is less than the LSD. In the latter case, we stop and make no further comparisons with $\bar{y}_{\text{smallest}}$. For our data, comparisons with $\bar{y}_{\text{smallest}}, \bar{y}_{1.}$ give the following results:

Comparison	Conclusion
$\bar{y}_{\text{largest}} - \bar{y}_{\text{smallest}} = \bar{y}_{5.} - \bar{y}_{1.} = .325$	>LSD; proceed
$\bar{y}_{\text{2nd largest}} - \bar{y}_{\text{smallest}} = \bar{y}_{4.} - \bar{y}_{1.} = .240$	>LSD; proceed
$\bar{y}_{\text{3rd largest}} - \bar{y}_{\text{smallest}} = \bar{y}_{3.} - \bar{y}_{1.} = .153$	>LSD; proceed
$\bar{y}_{\text{4th largest}} - \bar{y}_{\text{smallest}} = \bar{y}_{2.} - \bar{y}_{1.} = .118$	<LSD; stop

To summarize our results, we make the following diagram:

Agent 1 2 3 4 5

Those populations joined by the underline have means that are not significantly different from $\bar{y}_1$. Note that agents 3, 4, and 5 have sample differences with agent 1 that exceed LSD and hence are not underlined.

- c. We now make similar comparisons with $\bar{y}_{2\text{nd smallest}}$, $\bar{y}_2$. In this case, we use the procedure of part (b).

Comparison	Conclusion
$\bar{y}_3 - \bar{y}_2 = .207$	>LSD; proceed
$\bar{y}_4 - \bar{y}_2 = .122$	<LSD; stop

Agent 1 2 3 4 5

- d. Continue the comparisons with $\bar{y}_{3\text{rd smallest}}$, $\bar{y}_3$, in this example.

Comparison	Conclusion
$\bar{y}_5 - \bar{y}_3 = .172$	>LSD; proceed
$\bar{y}_4 - \bar{y}_3 = .087$	<LSD; stop

Agent 1 2 3 4 5

- e. Continue the comparisons with $\bar{y}_{4\text{th smallest}}$, $\bar{y}_4$, in this example.

Comparison	Conclusion
$\bar{y}_5 - \bar{y}_4 = .085$	<LSD; stop

Agent 1 2 3 4 5

- f. We can summarize steps (a) through (e) as follows:

Agent 1 2 3 4 5

Those populations not underlined by a common line are declared to have means that are significantly different according to the LSD criterion. Note that we can eliminate the third line from the top of part (f) since it is contained in the second line from the top. The revised summary of significant and nonsignificant results is

Agent 1 2 3 4 5

In conclusion, we have μ_1 , μ_2 , and μ_3 significantly less than μ_5 . Also, μ_3 and μ_4 are significantly greater than μ_1 .

Although the LSD procedure described in Example 9.6 may seem quite laborious, its application is quite simple. First, we run an analysis of variance. If

we reject the null hypothesis of equality of the population means, we compute the LSD for all pairs of sample means. When the sample sizes are the same, this difference is a single number for all pairs. We can use the stepwise procedure described in steps 5(a) through 5(f) of Example 9.6. We need not write down all those steps, only the summary lines. The final summary, as given in step 5(f), gives a handy visual display of the pairwise comparisons using Fisher's LSD.

Several remarks should be made concerning the LSD method for pairwise comparisons. First, there is the possibility that the overall F test in our analysis of variance is significant but that no pairwise differences are significant using the LSD procedure. This apparent anomaly can occur because the null hypothesis $H_0: \mu_1 = \mu_2 = \cdots = \mu_t$ for the F test is equivalent to the hypothesis that all possible comparisons (paired or otherwise) among the population means are zero. For a given set of data, the comparisons that are significant might not be of the form $\mu_i - \mu_j$, the form we are using in our paired comparisons.

Second, Fisher's LSD procedure can also be used to form a confidence interval for $\mu_i - \mu_j$. A $100(1 - \alpha)\%$ confidence interval has the form

$$(\bar{y}_i - \bar{y}_j) \pm \text{LSD}_{ij}$$

Third, when all sample sizes are the same, the LSD for all pairs is

$$t_{\alpha/2} \sqrt{\frac{2s_W^2}{n}}$$

Fisher's confidence interval

LSD for equal sample sizes

9.5 Tukey's W Procedure

We are aware of the major drawback of a multiple-comparison procedure with a controlled per-comparison error rate. Even when $\mu_1 = \mu_2 = \cdots = \mu_t$, unless α , the per-comparison error rate (such as with Fisher's unprotected LSD) is quite small, there is a high probability of declaring at least one pair of means significantly different when running multiple comparisons. To avoid this, other multiple-comparison procedures have been developed that control different error rates.

Tukey (1953) proposed a procedure that makes use of the **Studentized range distribution**. When more than two sample means are being compared, to test the largest and smallest sample means, we could use the test statistic

$$\frac{\bar{y}_{\text{largest}} - \bar{y}_{\text{smallest}}}{s_p \sqrt{1/n}}$$

where n is the number of observations in each sample and s_p is a pooled estimate of the common population standard deviation σ . This test statistic is very similar to that for comparing two means, but it does not possess a t distribution. One reason it does not is that we have waited to determine which two sample means (and hence population means) we would compare until we observed the largest and smallest sample means. This procedure is quite different from that of specifying $H_0: \mu_1 - \mu_2 = 0$, observing $\bar{y}_1$ and $\bar{y}_2$, and forming a t statistic.

The quantity

$$\frac{\bar{y}_{\text{largest}} - \bar{y}_{\text{smallest}}}{s_p \sqrt{1/n}}$$

follows a Studentized range distribution. We will not discuss the properties of this distribution, but will illustrate its use in Tukey's multiple-comparison procedure.

Studentized range distribution

Tukey's W Procedure
 W
 $q_\alpha(t, v)$

experimentwise error rate

1. Rank the t sample means.
2. Two population means μ_i and μ_j are declared different if

$$|\bar{y}_i - \bar{y}_j| \geq W$$

where

$$W = q_\alpha(t, v) \sqrt{\frac{s_W^2}{n}}$$

s_W^2 is the mean square within samples based on v degrees of freedom, $q_\alpha(t, v)$ is the **upper-tail critical value of the Studentized range** for comparing t different populations, and n is the number of observations in each sample. A discussion follows showing how to obtain values of $q_\alpha(t, v)$ from Table 10 in the Appendix.

3. The error rate that is controlled is an **experimentwise error rate**. Thus, the probability of observing an experiment with one or more pairwise comparisons falsely declared to be significant is specified at α .

We can obtain values of $q_\alpha(t, v)$ from Table 10 in the Appendix. Values of v are listed along the left column of the table with values of t across the top row. Upper-tail values for the Studentized range are then presented for $\alpha = .05$ and $.01$. For example, in comparing 10 population means based on 9 degrees of freedom for s_W^2 , the $.05$ upper-tail critical value of the Studentized range is $q_{.05}(10, 9) = 5.74$.

EXAMPLE 9.7

Refer to the data of Example 9.3. Use Tukey's W procedure with $\alpha = .05$ to make pairwise comparisons among the five population means.

Solution Step 1 is to rank the sample means from smallest to largest, to produce

Agent	1	2	3	4	5
$\bar{y}_i$	1.175	1.293	1.328	1.415	1.500

For the experiment described in Example 9.6, we have

- $t = 5$ (we are making pairwise comparisons among five means)
- $v = 25$ (s_W^2 had degrees of freedom equal to df_{Error} in the AOV)
- $\alpha = .05$ (we specified α_E , the experimentwise error rate at $.05$)
- $n = 6$ (there were six plots randomly assigned to each of the agents)

We find in Table 10 of the Appendix that

$$q_\alpha(t, v) = q_{.05}(5, 25) \approx 4.158$$

The absolute value of each difference in the sample means $|\bar{y}_i - \bar{y}_j|$ must then be compared to

$$W = q_\alpha(t, v) \sqrt{\frac{s_W^2}{n}} = 4.158 \sqrt{\frac{.0153}{6}} = .2100$$

By substituting W for LSD, we can use the same stepwise procedure for comparing sample means that we used in step 5 of the solution to Example 9.6. Having

ranked the sample means from low to high, we compare against $\bar{y}_{\text{smallest}}$, which is $\bar{y}_1$, as follows:

Comparison	Conclusion
$\bar{y}_{\text{largest}} - \bar{y}_{\text{smallest}} = \bar{y}_5 - \bar{y}_1 = .325$	$> W$; proceed
$\bar{y}_{\text{2nd largest}} - \bar{y}_{\text{smallest}} = \bar{y}_4 - \bar{y}_1 = .240$	$> W$; proceed
$\bar{y}_{\text{3rd largest}} - \bar{y}_{\text{smallest}} = \bar{y}_3 - \bar{y}_1 = .153$	$< W$; stop

To summarize our results we make the following diagram:

Agent 1 2 3 4 5

Comparison with $\bar{y}_{\text{2nd smallest}}$, which is $\bar{y}_2$, yields

Comparison	Conclusion
$\bar{y}_5 - \bar{y}_2 = .207$	$< W$; stop

Agent 1 2 3 4 5

Similarly, comparisons of $\bar{y}_5$ with $\bar{y}_3$ and $\bar{y}_4$ yield

Agent 1 2 3 4 5

Combining our results, we obtain

Agent 1 2 3 4 5

which simplifies to

Agent 1 2 3 4 5

All populations not underlined by a common line have population means that are significantly different from each other; that is, μ_4 and μ_5 are significantly larger than μ_1 . No other pairs of means are significantly different.

By examining the multiple-comparison summaries using the least significant difference (Example 9.6) and Tukey's W procedure (Example 9.7), we see that Tukey's procedure is more conservative (declares fewer significant differences) than the LSD procedure. For example, when applying Tukey's procedure to the data in Example 9.6, we found that μ_3 is no longer significantly larger than μ_1 . Similarly, μ_2 and μ_3 are no longer significantly less than μ_5 . The explanation for this is that although both procedures have an experimentwise error rate, we have shown the per-comparison error rate of the protected LSD method to be larger than that for Tukey's W procedure.

A limitation of Tukey's procedure is the requirement that all the sample means are based on the same number of data values. There are no available tables for the case in which the sample sizes are unequal. If the sample sizes are only